



Project Title VAMC Durham - Replace Chillers for AHU 1 & 2			Project Number VA Project No. 558-22-172 AG Project No. 0122096	
Location DURHAM VA			Drawing Number G-001	
Issue Date 3/26/2026	Checked EJL	Drawn SJ		

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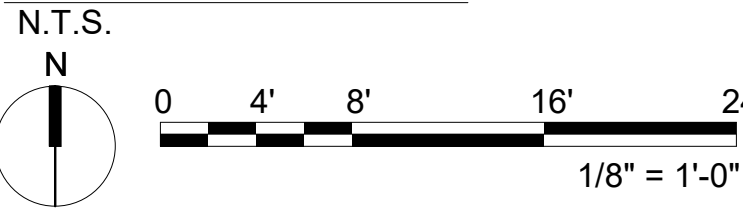
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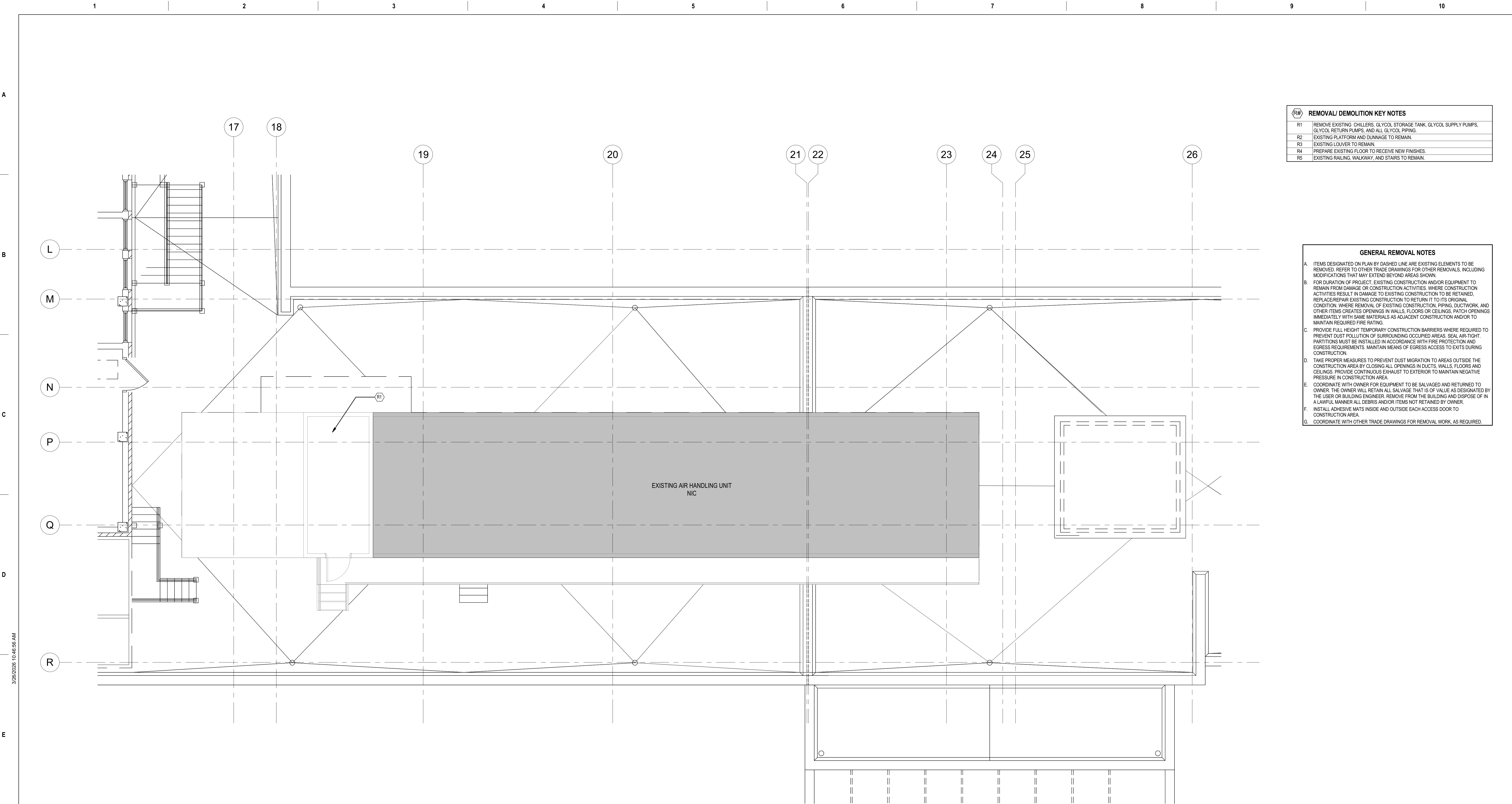
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1. TOP OF EXISTING ROOF IS 131.09' ABOVE DATUM (386.00').
2. TOP OF EXISTING EQUIPMENT PLATFORM IS 2.17' ABOVE TOP OF EXISTING ROOF (133.25' ABOVE DATUM).
3. CONTRACTOR TO VERIFY DIMENSIONS NOTED AS V.I.F. PRIOR TO STEEL FABRICATION. COORDINATE EQUIPMENT PLATFORM WIDTH WITH FINAL MECHANICAL EQUIPMENT SELECTION. IF PLATFORM WIDTH EXTENSION IS NOT SUFFICIENT FOR FINAL MECHANICAL EQUIPMENT SELECTION, COR IS TO BE NOTIFIED AND REDESIGN AND/OR REINFORCEMENT OF THE PLATFORM MAY BE REQUIRED IN THAT CASE.
4. STEEL TO BE PAINTED TO MATCH EXISTING.

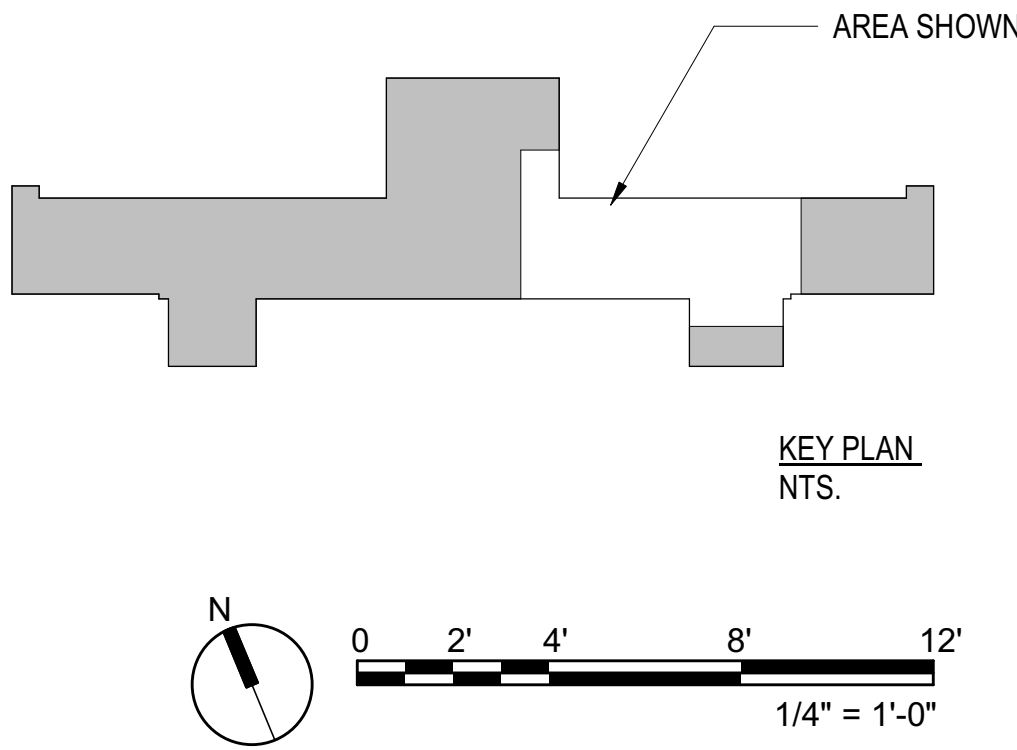
- 1 EXISTING CONCRETE ON METAL DECK PLATFORM IS COMPOSED OF 1 1/2" TYPE 'B', METAL ROOF DECK WITH 2 1/2" NORMAL WEIGHT CONCRETE (4" TOTAL).
- 2 EXISTING METAL GRATING WALKWAY PLATFORM.
- 3 REMOVE PAINT AND CLEAN STEEL TO SPCC 2 STANDARD IN AREAS WHERE FIELD WELDING IS TO BE PERFORMED. REPAIR PAINT AFTER WELDING HAS BEEN COMPLETED.
- 4 ADD CONCRETE ON METAL DECK PLATFORM COMPOSED OF 1 1/2" TYPE 'B', METAL ROOF DECK WITH 2 1/2" NORMAL WEIGHT CONCRETE (4" TOTAL). ATTACH TO SUPPORTING STRUCTURE WITH #12 SCREWS IN A 36" PATTERN WITH #10 SELF-DRILLING SCREWS EQUALLY SPACED.
- 5 REMOVE LOOSE CONCRETE AND DEBRIS FROM WORK AREA. PATCH AND LEVEL CONCRETE AND APPLY EXTERIOR EXPOSURE RATED SEALANT.
- 6 CHILLER, REFER TO MECHANICAL DRAWINGS, BASIS OF DESIGN WEIGHT 12,010 LBS.
- 7 DRY CLEAN METAL GYLOID STORAGE TANK, REFER TO MECHANICAL DRAWINGS, BASIS OF DESIGN WEIGHT 10,890 LBS. THROUGH BOLT STEEL SUPPORT FRAME THROUGH DECK WITH 3/4" DIA. BOLTS AT 12" O.C.
- 8 PUMP, REFER TO MECHANICAL DRAWINGS, BASIS OF DESIGN WEIGHT 650 LBS. EACH.
- 9 GYLOID FEED SYSTEM, REFER TO MECHANICAL DRAWINGS, BASIS OF DESIGN WEIGHT 490 LBS.
- 10 EXPANSION TANK, REFER TO MECHANICAL DRAWINGS, BASIS OF DESIGN WEIGHT 300 LBS.
- 11 AIR SEPARATOR, REFER TO MECHANICAL DRAWINGS, BASIS OF DESIGN WEIGHT 570 LBS.
- 12 OUTDOOR UNIT, REFER TO MECHANICAL DRAWINGS, BASIS OF DESIGN WEIGHT 290 LBS.
- 13 TRANSFORMER, REFER TO ELECTRICAL DRAWINGS, BASIS OF DESIGN WEIGHT 225 LBS. PROVIDE 4" THICK HOUSEKEEPING PAD, REFER TO DETAIL 1-S/301.

[illegible]



R#	REMOVAL/ DEMOLITION KEY NOTES
R1	REMOVE EXISTING CHILLERS, GLYCOL STORAGE TANK, GLYCOL SUPPLY PUMPS, GLYCOL RETURN PUMPS, AND ALL GLYCOL PIPING.
R2	EXISTING PLATFORM AND DUNNAGE TO REMAIN.
R3	EXISTING LOUVER TO REMAIN.
R4	PREPARE EXISTING FLOOR TO RECEIVE NEW FINISHES.
R5	EXISTING RAILING, WALKWAY, AND STAIRS TO REMAIN.

- GENERAL REMOVAL NOTES**
- A. ITEMS DESIGNATED ON PLAN BY DASHED LINE ARE EXISTING ELEMENTS TO BE REMOVED. REFER TO OTHER TRADE DRAWINGS FOR OTHER REMOVALS, INCLUDING MODIFICATIONS THAT MAY EXTEND BEYOND AREAS SHOWN.
- B. FOR DURATION OF PROJECT, EXISTING CONSTRUCTION AND/OR EQUIPMENT TO REMAIN FROM DAMAGE OR CONSTRUCTION ACTIVITIES, WHERE CONSTRUCTION ACTIVITIES RESULT IN DAMAGE TO EXISTING CONSTRUCTION TO BE RETAINED, REPLACE/REPAIR EXISTING CONSTRUCTION TO RETURN IT TO ITS ORIGINAL CONDITION, WHERE REMOVAL OF EXISTING CONSTRUCTION, PIPING, DUCTWORK, AND OTHER ITEMS CREATES OPENINGS IN WALLS, FLOORS OR CEILINGS, PATCH OPENINGS IMMEDIATELY WITH SAME MATERIALS AS ADJACENT CONSTRUCTION AND/OR TO MAINTAIN REQUIRED FIRE RATING.
- C. PROVIDE FULL HEIGHT TEMPORARY CONSTRUCTION BARRIERS WHERE REQUIRED TO PREVENT DUST POLLUTION OF SURROUNDING OCCUPIED AREAS. SEAL AIR-TIGHT. PARTITIONS MUST BE INSTALLED IN ACCORDANCE WITH FIRE PROTECTION AND EGRESS REQUIREMENTS. MAINTAIN MEANS OF EGRESS ACCESS TO EXITS DURING CONSTRUCTION.
- D. TAKE PROPER MEASURES TO PREVENT DUST MIGRATION TO AREAS OUTSIDE THE CONSTRUCTION AREA BY CLOSING ALL OPENINGS IN DUCTS, WALLS, FLOORS AND CEILINGS. PROVIDE CONTINUOUS EXHAUST TO EXTERIOR TO MAINTAIN NEGATIVE PRESSURE IN CONSTRUCTION AREA.
- E. COORDINATE WITH OWNER FOR EQUIPMENT TO BE SALVAGED AND RETURNED TO OWNER. THE OWNER WILL RETAIN ALL SALVAGE THAT IS OF VALUE AS DESIGNATED BY THE USER OR BUILDING ENGINEER. REMOVE FROM THE BUILDING AND DISPOSE OF IN A LAWFUL MANNER ALL DEBRIS AND/OR ITEMS NOT RETAINED BY OWNER.
- F. INSTALL ADHESIVE MATS INSIDE AND OUTSIDE EACH ACCESS DOOR TO CONSTRUCTION AREA.
- G. COORDINATE WITH OTHER TRADE DRAWINGS FOR REMOVAL WORK, AS REQUIRED.



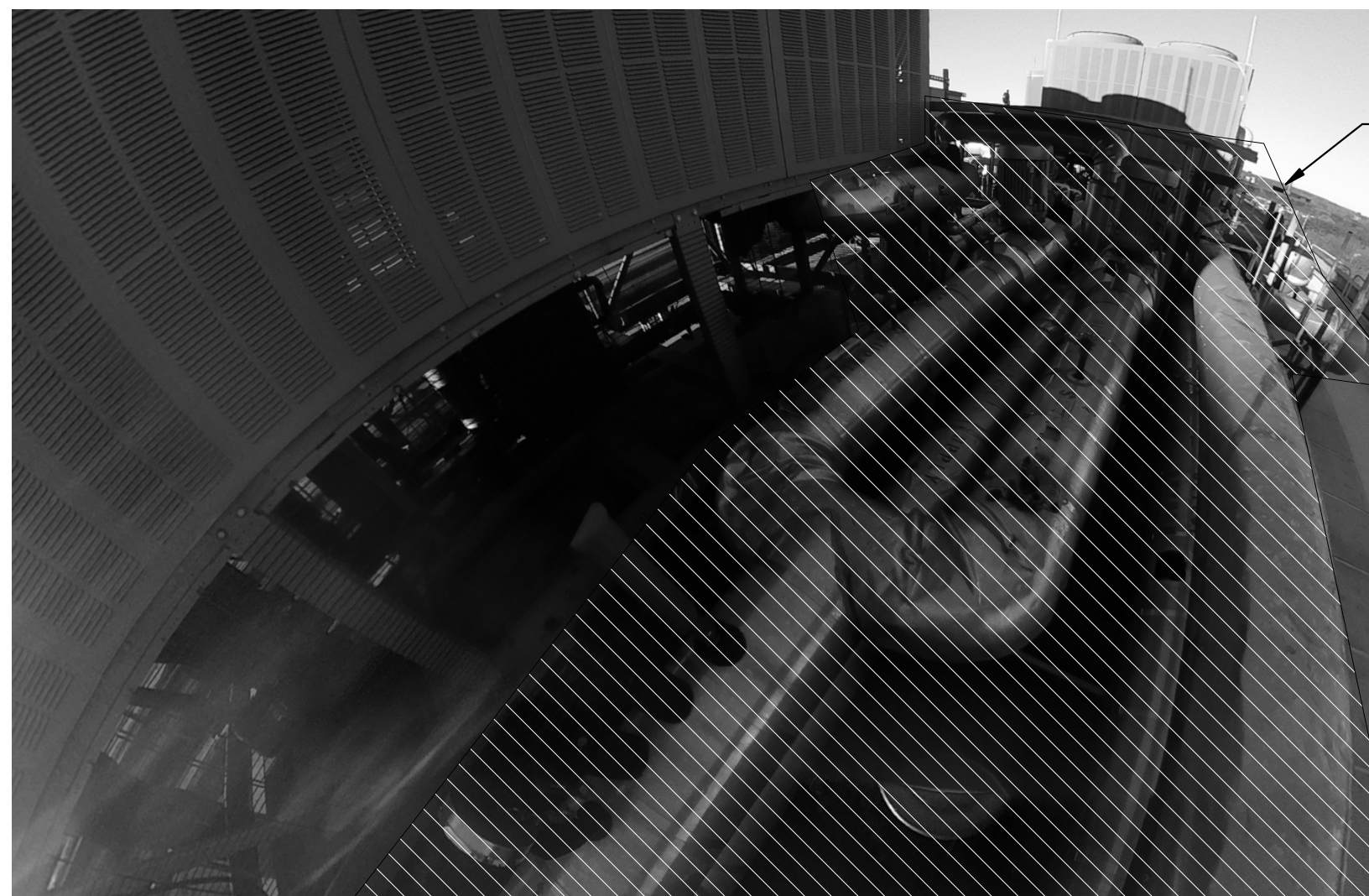
1 DEMO - ROOF PLAN B
AD-102 SCALE: 1/4" = 1'-0"

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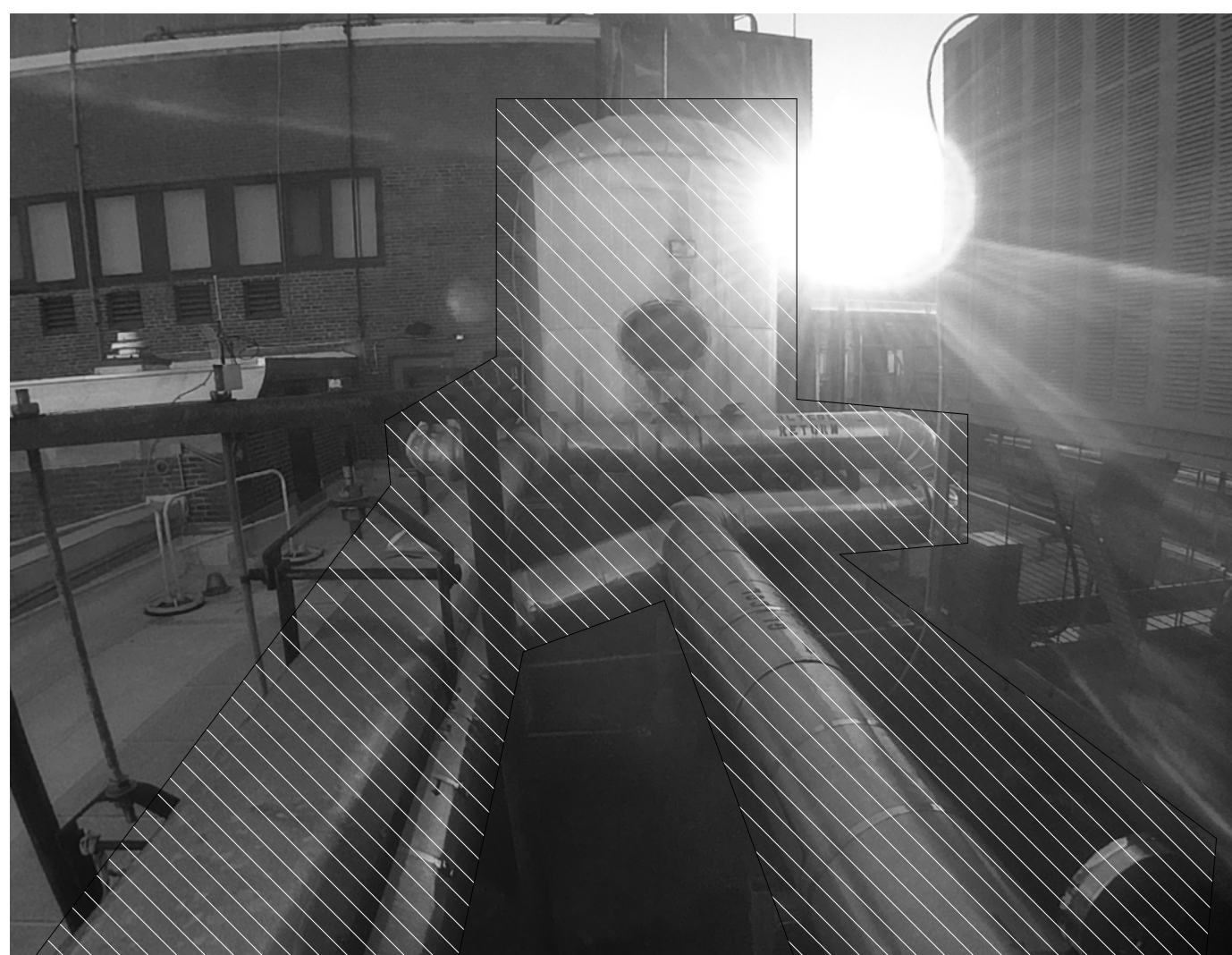
Revisions		CONSULTANT DELTA ENGINEERS, ARCHITECTS, & SURVEYORS 8401 Connecticut Avenue, Suite 350 Chevy Chase, MD 20815 Tel: 301.718.0080 Fax: 301.718.9520 Email: mail@delta-eas.com www.delta-eas.com	ARCHITECT/ENGINEER OF RECORD Above Group ENGINEERING. DESIGN. CONSULTING. www.abovegroupinc.com COA/ICA Lic. No. 31120 305 East Dr., Suite H, Melbourne, Florida 32904 PH: 321.345.9026		STAMP 	Office of Construction and Facilities Management U.S. Department of Veterans Affairs	Drawing Title PARTIAL ROOF DEMOLITION PLAN Approved:	Phase FINAL CONSTRUCTION DOCUMENTS FULLY SPRINKLERED	Project Title VAMC Durham - Replace Chillers for AHU 1 & 2 Location DURHAM VA Issue Date 3/26/2026	Project Number VA Project No. 558-22-172 AG Project No. 0122096 Building Number 1 Drawing Number AD-102
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2 EXISTING CHILLED WATER PUMPS WITH DISCONNECTS
N.T.S.



3 EXISTING CHILLED WATER PUMPS
N.T.S.



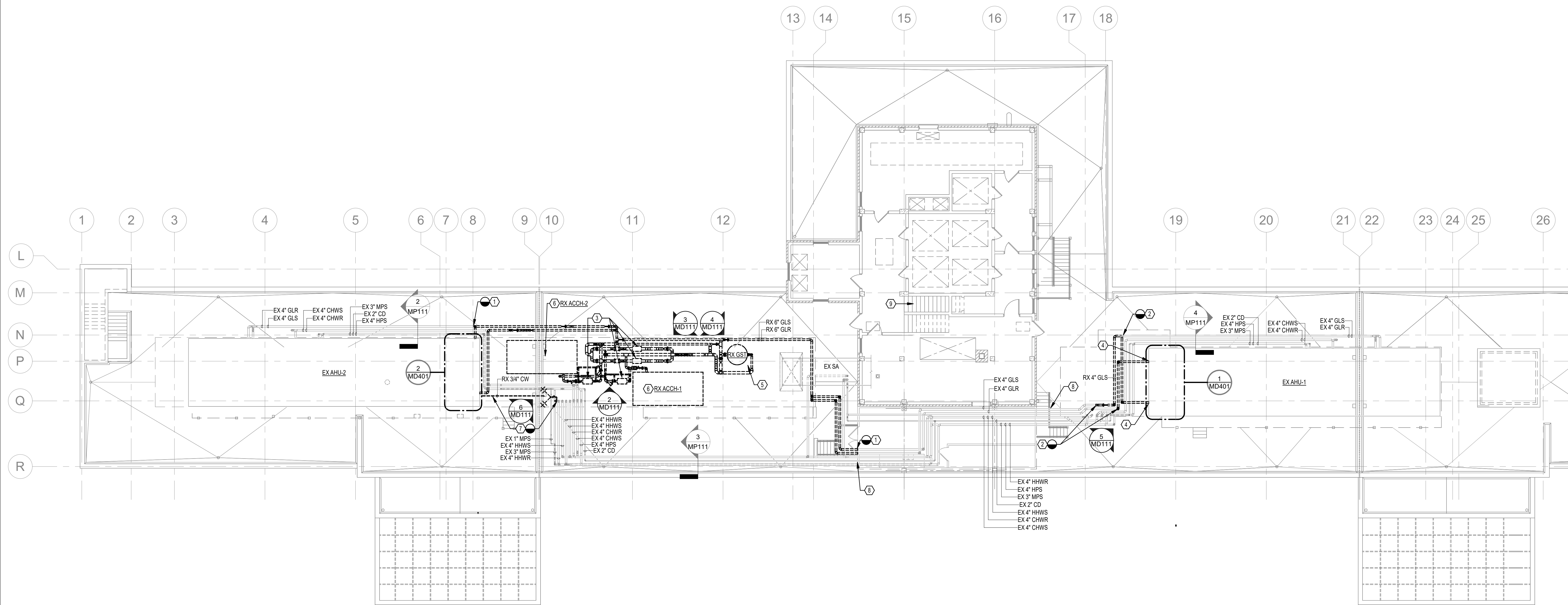
4 EXISTING GLYCOL STORAGE TANK
N.T.S.

DRAWING NOTES:

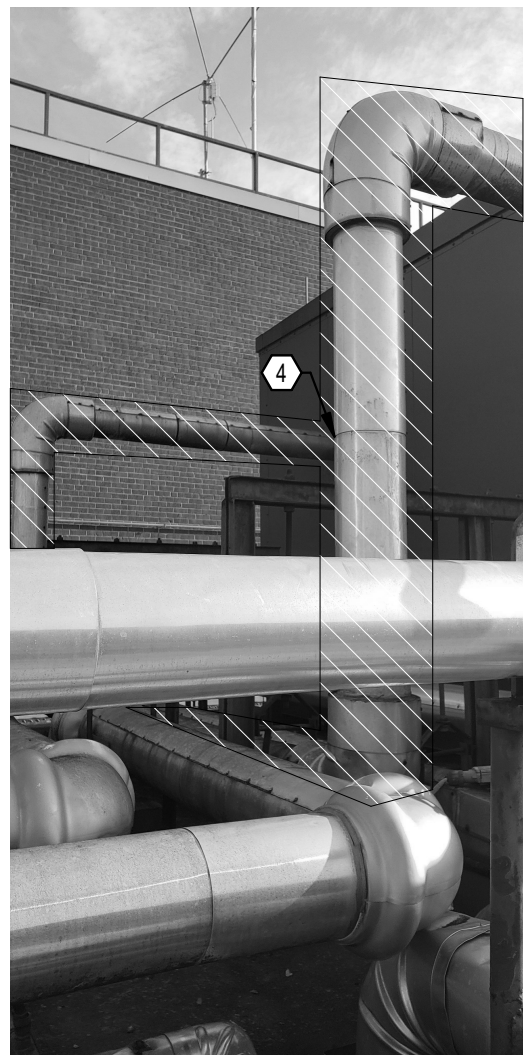
1. REFER TO MD01 FOR PIPING DIAGRAM OF EXISTING CHILLER ARRANGEMENT.
2. REMOVE ALL INSULATION FROM EX GLS AND GLR PIPING AND ACCESSORIES. INSPECT ALL PIPING FOR LEAKS AND DEFICIENCIES. PREP PIPING FOR NEW INSULATION.
3. ALL REMOVED EQUIPMENT SHALL BE CRANED OFF OF ROOF TO PARKING LOT.

CODED NOTES:

- 1 REMOVE EXISTING GLS AND REMOVE EXISTING GLR BACK TO LOCATION SHOWN
- 2 REMOVE EXISTING GLR BACK TO LOCATION SHOWN
- 3 REMOVE EXISTING PUMPS, ALL GLS AND GLR PIPING, AND ALL ACCESSORIES BACK TO LOCATION SHOWN. REMOVE EXISTING PIPE SUPPORTS AND PUMP ENCLOSURES. COORDINATE REMOVAL OF DISCONNECTS WITH ELECTRICAL CONTRACTOR.
- 4 REMOVE EXISTING GLR PIPING BACK INTO AHU. PATCH AND SEAL AHU PENETRATIONS WATER TIGHT. REMOVE EXISTING GLR PIPING TO LOCATION SHOWN ON PLAN.
- 5 REMOVE EXISTING GLYCOL STORAGE TANK AND ALL ASSOCIATED PIPING AND ACCESSORIES BACK TO LOCATION SHOWN ON DRAWING.
- 6 REMOVE EXISTING CHILLER AND ALL ASSOCIATED PIPING AND ACCESSORIES BACK TO LOCATION SHOWN ON DRAWING.
- 7 REMOVE EXISTING CW, HOSE CONNECTIONS, BALL VALVE, PIPING, AND GLOBE VALVE. EXISTING PIPING TO HOSE BBB IN MECHANICAL ROOM IS CRACKED, NOT INSULATED, AND ABANDONED IN PLACE.
- 8 PIPING ROUTES UNDER MECHANICAL PENTHOUSE FROM THIS POINT TILL NEXT INDICATED LOCATION.
- 9 NO ELEVATOR ACCESS TO PENTHOUSE THAT ACCESSES THE ROOF. ALL MATERIALS NEED TO BE CARRIED UP AND DOWN STAIRCASE OR CRANED TO ROOF.



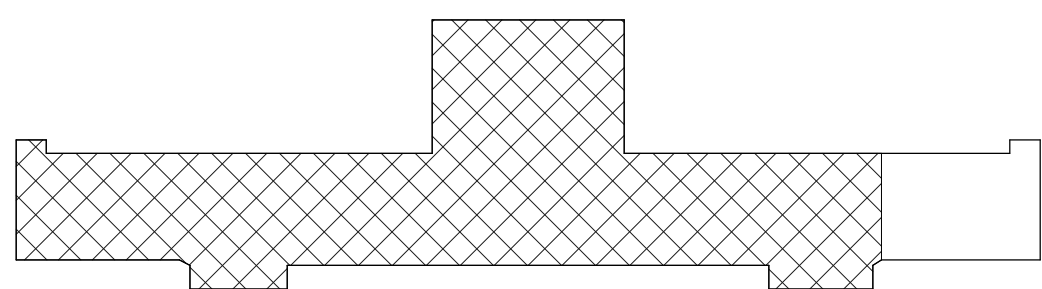
1 ROOF PLAN - MECHANICAL - DEMOLITION
3/32" = 1'-0"



5 EXISTING AHU-1 GLR REMOVAL
N.T.S.



6 COLD WATER CONNECTION REMOVAL
N.T.S.



KEY PLAN - ROOF

N.T.S.
0 8' 16' 24' 32'
3/32" = 1'-0"

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Above Group
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JOSHUA CALHOUN, P.E.
PA License #PE093054

Office of
Construction
and Facilities
Management

VA U.S. Department of
Veterans Affairs

Drawing Title

MECHANICAL ROOF PLAN -
DEMOLITION

Approved:

Phase

FINAL CONSTRUCTION
DOCUMENTS

FULLY SPRINKLERED

Project Title

VAMC Durham - Replace
Chillers for AHU 1 & 2

Location

DURHAM VA

Issue Date

03/26/2026

Checked

JMC

Drawn

AJZ

Project Number

VA Project No. 558-22-172

AG Project No. 0122096

Building Number

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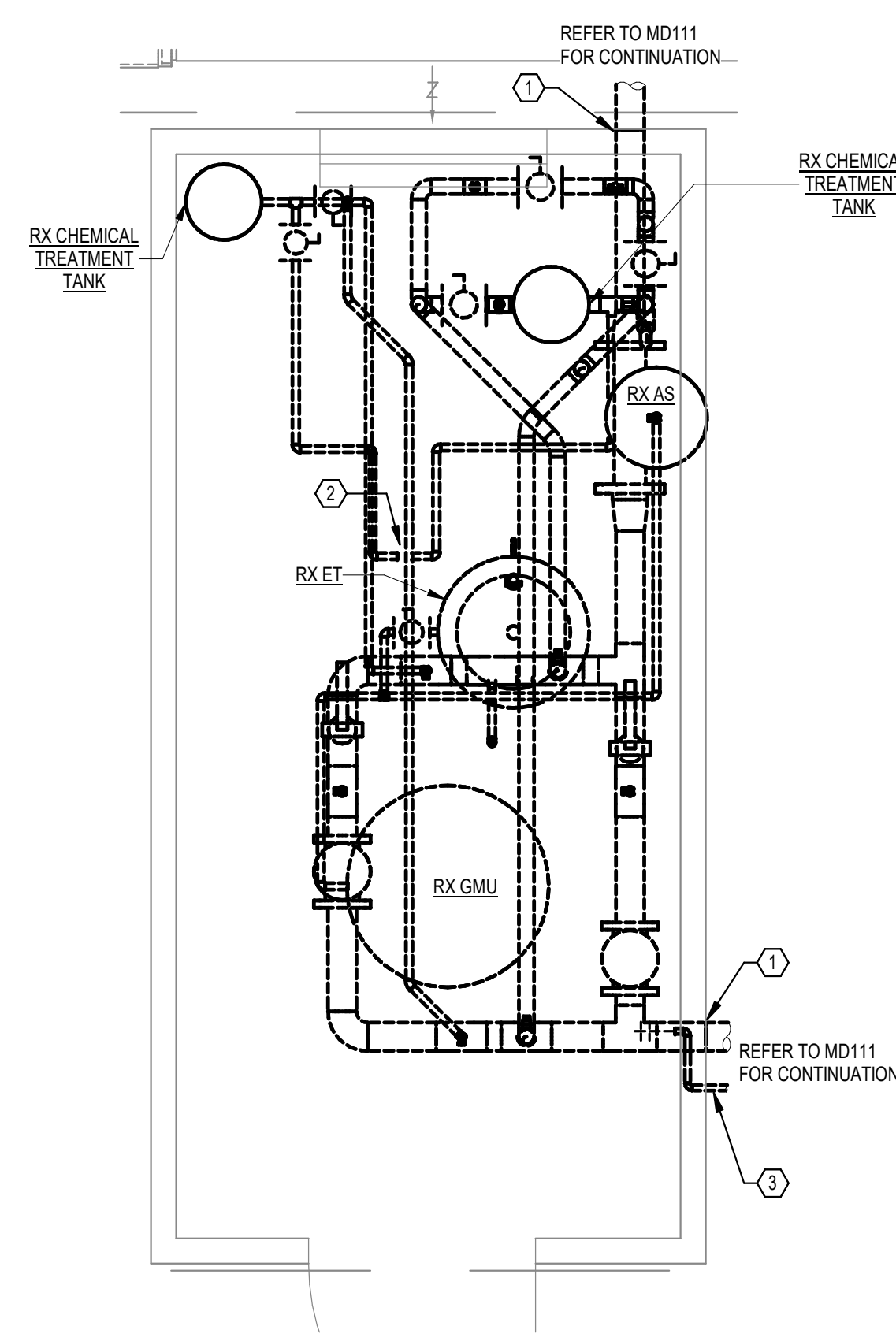
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MD111

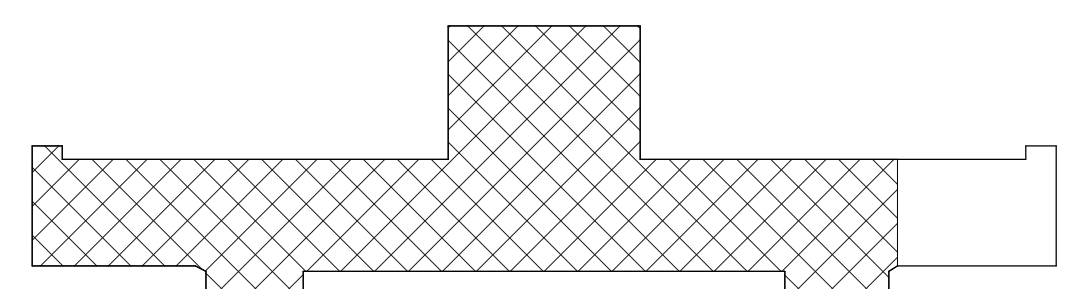
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VA FORM 08 - 6231



2 AHU-2 MECHANICAL ROOM - MECHANICAL - DEMOLITION
1/2" = 1'-0"

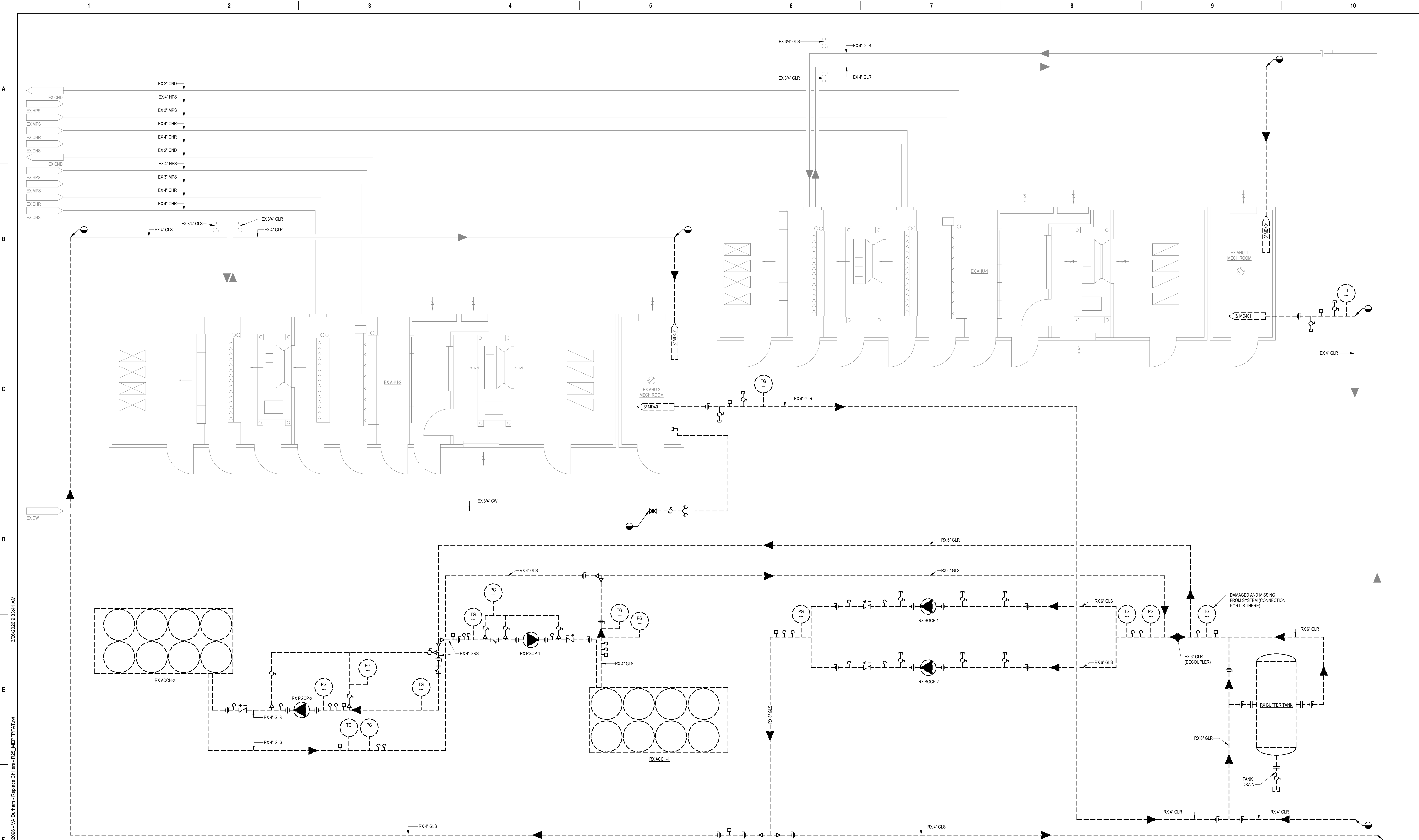


KEY PLAN - ROOF
N.T.S.

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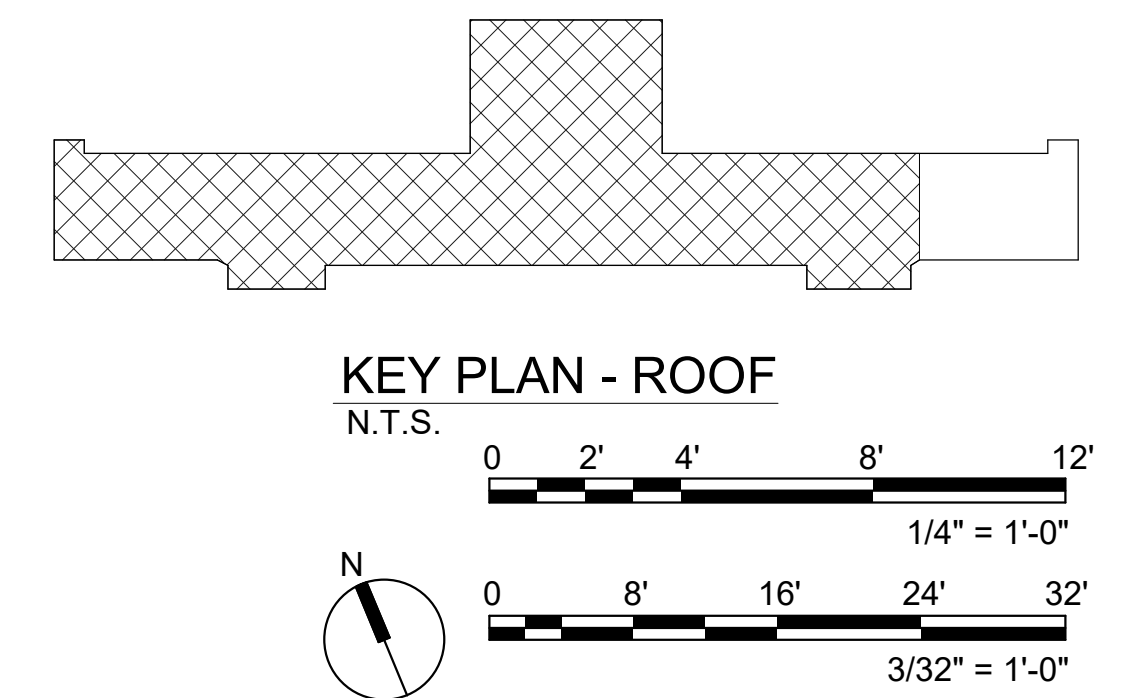
0 1' 2' 4' 6'

1/2" = 1'-0"

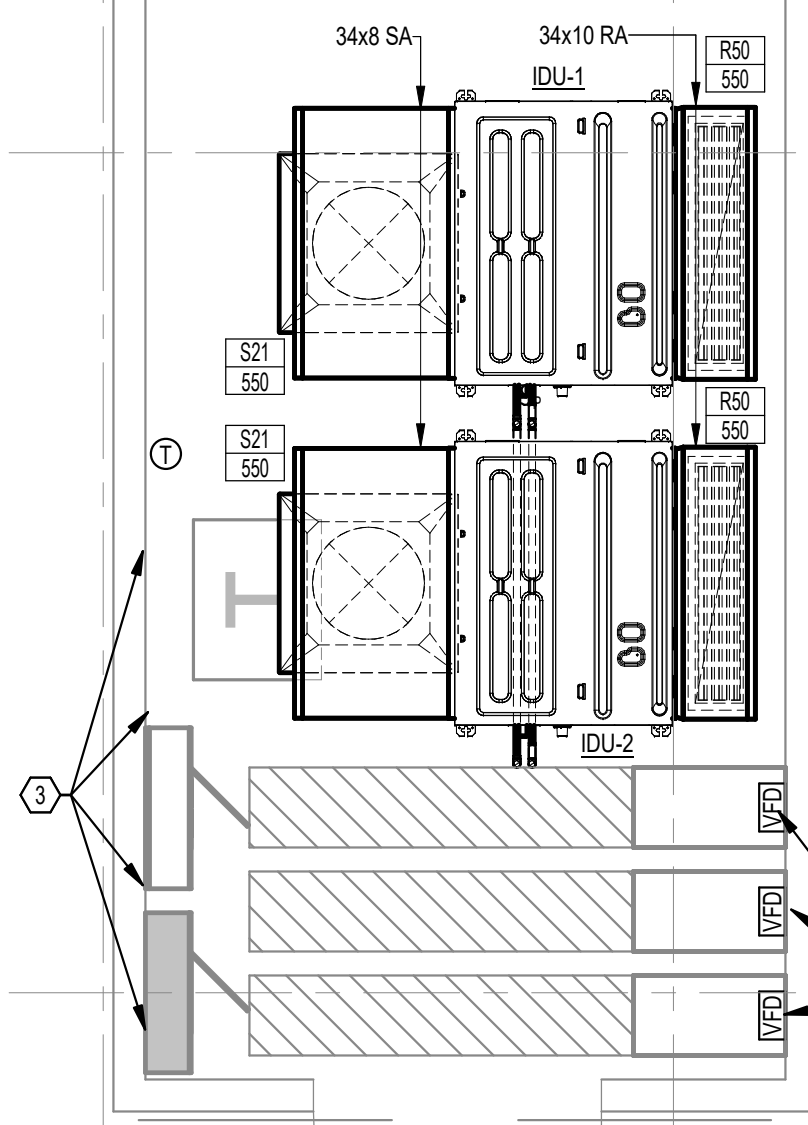


1 CHILLER P&ID - DEMOLITION
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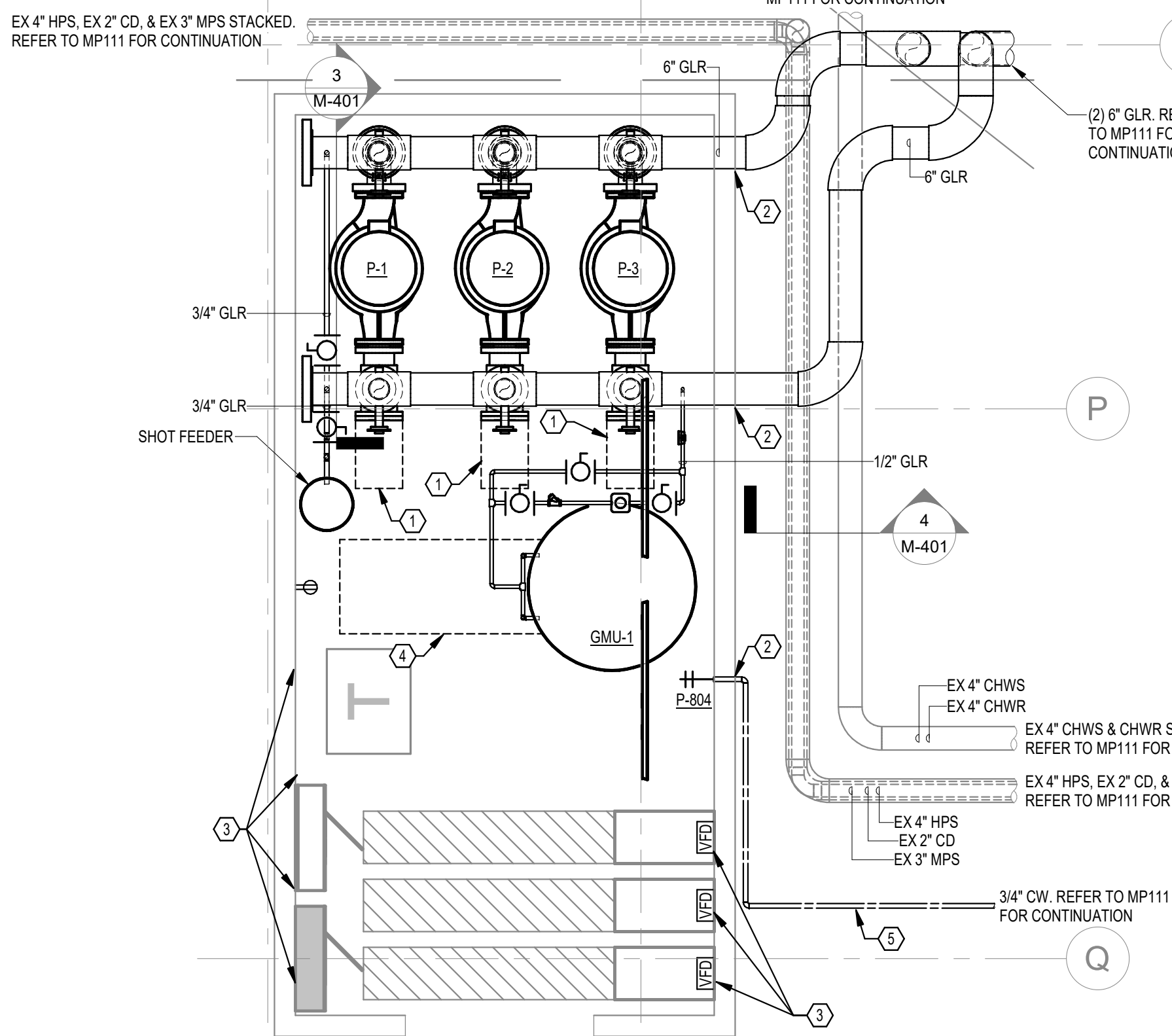
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Revisions		Date			COMMONWEALTH OF PENNSYLVANIA JOSHUA CALHOUN PROFESSIONAL ENGINEER PE093054 SBA SMALL BUSINESS ALLIANCE VERIFIED CREDENTIAL CERTIFICATION JOSHUA CALHOUN, P.E. PA License #PE093054	U.S. Department of Veterans Affairs	Approved:	FULLY SPRINKLERED	Location DURHAM VA	Issue Date 03/26/2026	Checked JMC	Drawn AJZ	Drawing Number MD601



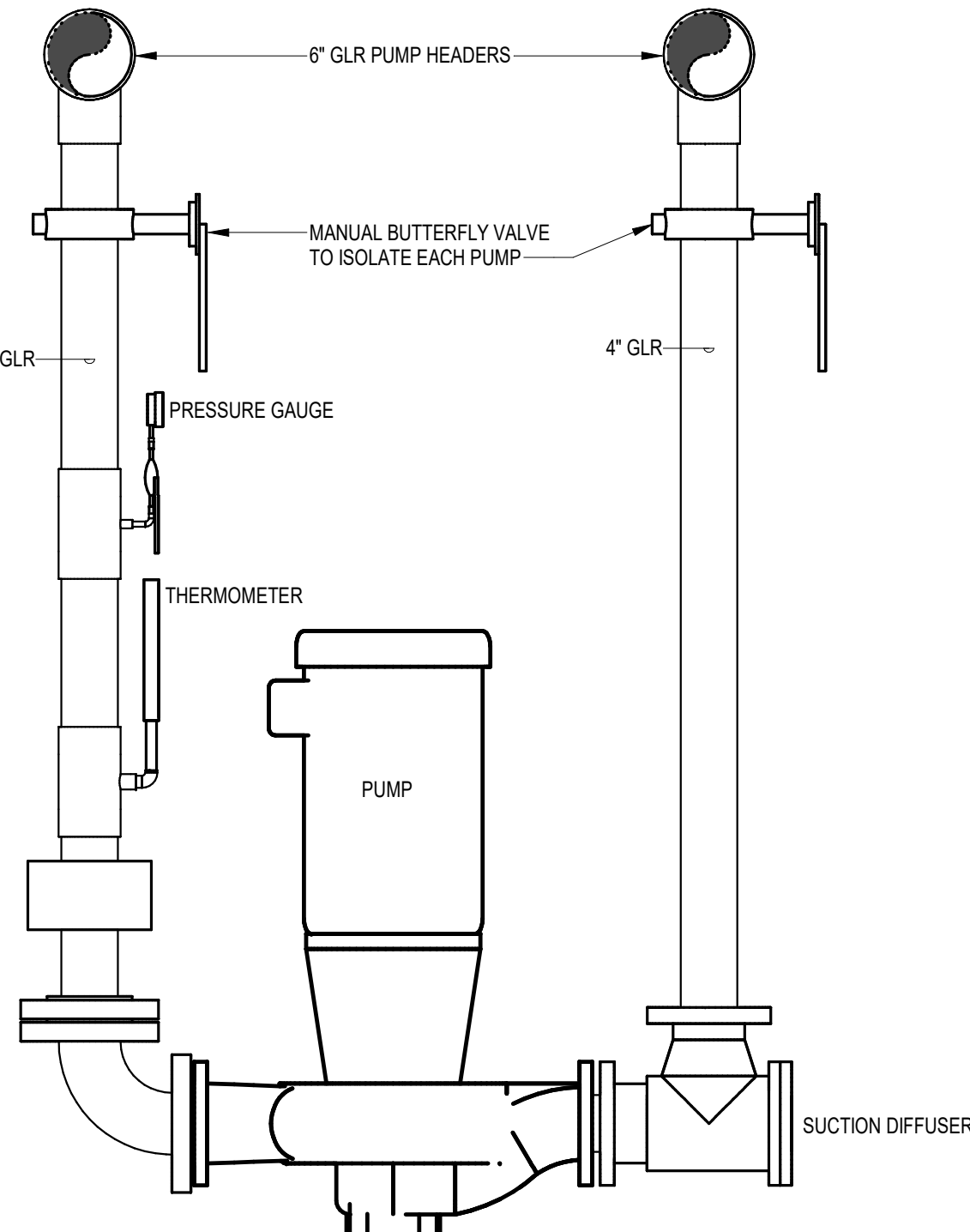
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			Above Group ENGINEERING. DESIGN. CONSULTING. www.abovegroupinc.com COA/CA Lic. No. 31120 305 East Dr., Suite H, Melbourne, Florida 32904 PH: 321.345.9026	COMMONWEALTH OF PENNSYLVANIA PROFESSIONAL ENGINEER JOSHUA CALHOUN PA License #PE093054 JOSHUA CALHOUN, P.E. PA License #PE093054		MECHANICAL ROOF PLAN - NEW WORK	FINAL CONSTRUCTION DOCUMENTS	VAMC Durham - Replace Chillers for AHU 1 & 2	VA Project No. 558-22-172 AG Project No. 0122096		
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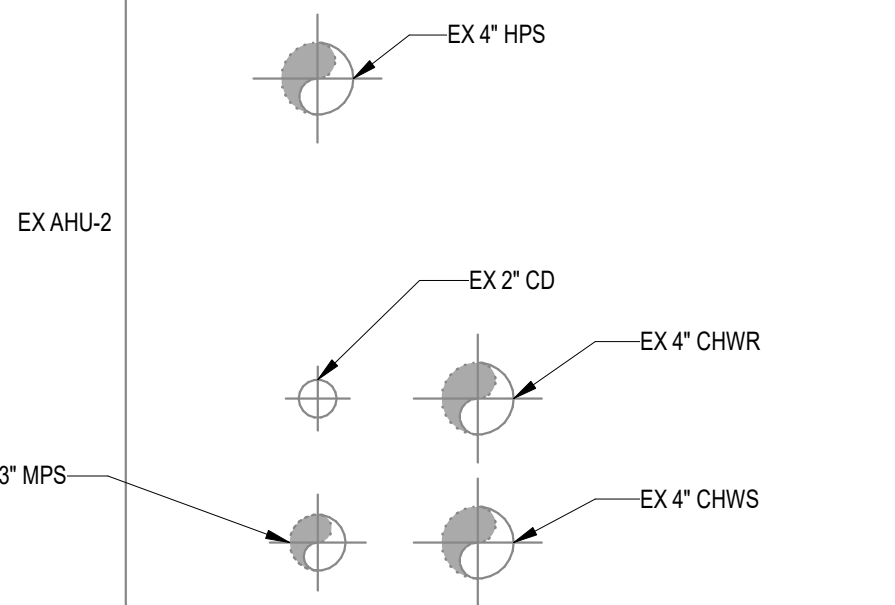
1 AHU-2
1/2" = 1'-0"



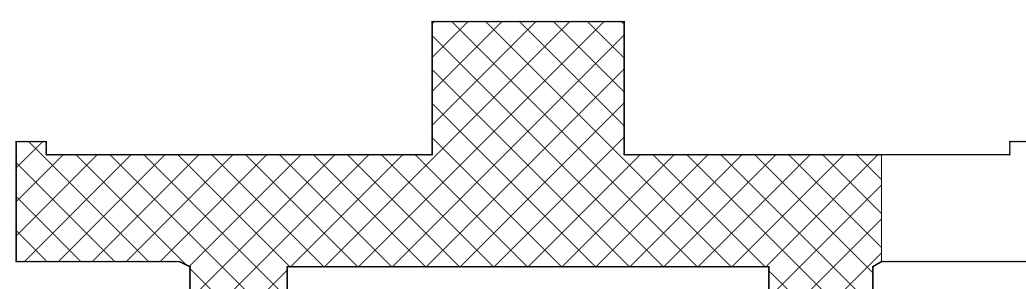
2 AHU-2
1/2" = 1'-0"



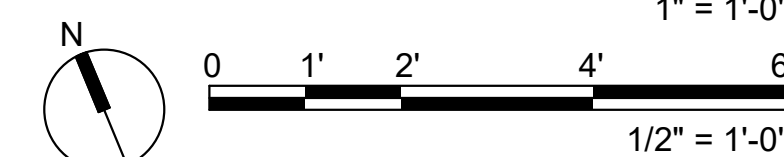
3 VERTICAL PUMP SECTION
1" = 1'-0"



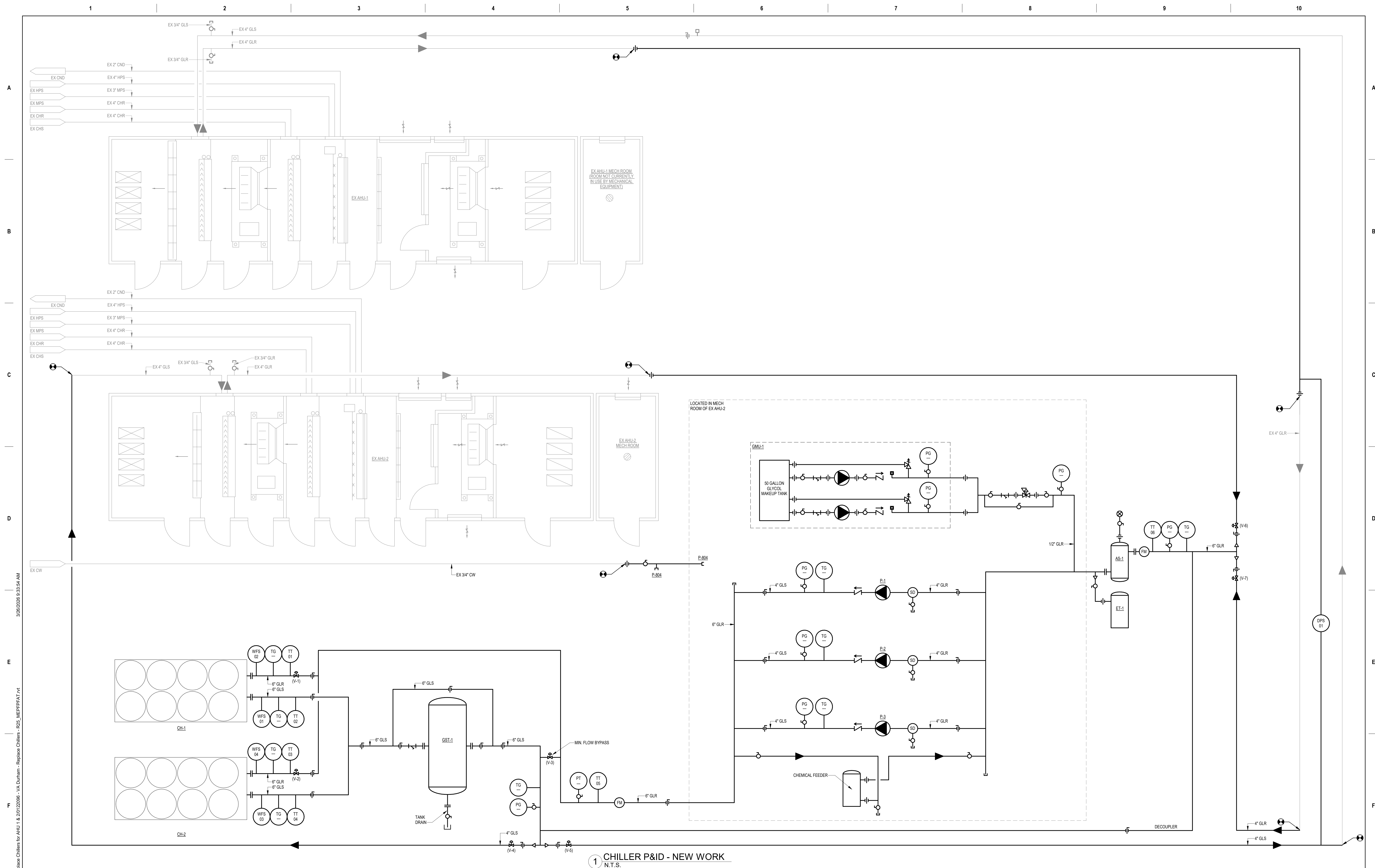
4 EX PIPING BETWEEN EX AHU-2 & EX CHILLER
1" = 1'-0"



KEY PLAN - ROOF
NTS

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Office of
Construction
and Facilities
Management

Drawing Title

SEQUENCE OF OPERATION

GENERAL:

- REFER TO MBD/FOR ALL SENSOR LOCATIONS AND CONTROL POINTS.
- THE CHILLED WATER PLANT CONSISTS OF TWO (2) 117 TON AIR COOLED CHILLERS. THE CHILLED WATER SYSTEM SHALL BE PROVIDED WITH (3) CHILLED WATER PUMPS FOR N+1 REDUNDANCY. EACH SERVED BY A SEPARATE VFD.
- A 2-WAY MODULATING BYPASS VALVE (V-3) SHALL BE PROVIDED TO ENSURE CHILLER PACKAGE MINIMUM FLOW.
- THE CHILLED WATER SYSTEM SHALL OPERATE BY VARIABLE PRIMARY PUMPING.
- A DIRECT DIGITAL CONTROL (DDC) SYSTEM SHALL BE PROVIDED THAT INTERFACES WITH THE CHILLER FACTORY CONTROLS AND INTEGRATES WITH THE EXISTING AHU CONTROLS TO PROVIDE FULL REMOTE CONTROL AND ALL INFORMATION FOR MONITORING AVAILABLE IN THE CHILLER CONTROLLER.
- CHILLED WATER PLANT SHALL OPERATE ONLY WHEN THE EXISTING BUILDING MAIN CHILLED WATER LOOP IS DISABLED OR INCAPABLE OF PROVIDING SUFFICIENT CHILLED WATER TO THE HVAC EQUIPMENT WITHIN THIS SCOPE OF WORK. WHEN THE DDC SYSTEM DETECTS LEAVING AIR TEMPERATURES GREATER THAN 2°F THAN ABOVE SETPOINT (42°F, ADJUSTABLE) OR RECEIVES A SIGNAL THAT THE MAIN BUILDING CHILLED WATER SYSTEM IS OFFLINE, THE DDC SYSTEM SHALL OPEN MOTORIZED ISOLATION VALVES V-6 AND V-7, AND SHALL ENABLE THE EMERGENCY CHILLED WATER PLANT.
- WHEN THE SYSTEM OPERATION OF THE MAIN BUILDING CHILLED WATER LOOP HAS RETURNED TO NORMAL OPERATION, THE DDC SYSTEM SHALL DISABLE THE EMERGENCY CHILLED WATER PLANT. SHALL CLOSE MOTORIZED ISOLATION VALVES V-6 AND V-7.

SYSTEM OPERATION:

- THE CHILLED WATER IS SIZED FOR THE FULL COOLING LOAD OF ASSOCIATED CHILLED WATER SYSTEM, GIVING THE SYSTEM N+1 REDUNDANCY WITH AN ADDITIONAL CIRCUIT. THEREFORE, ONLY ONE CHILLER AND TWO PUMPS SHALL OPERATE AT A TIME. THE CHILLERS SHALL BE CONFIGURED TO TRADE OFF OPERATION TO EVEN OUT RUNTIME. OPERATION SHALL BE AS FOLLOWS:
 - WEEK ONE: CH1 & P-1 & P-2 SHALL BE IN OPERATION AND CH2 & P-3 SHALL BE DE-ENERGIZED (ADJUSTABLE).
 - WEEK TWO: CH2 & P-2 & P-3 SHALL BE IN OPERATION AND CH1 & P-1 SHALL BE DE-ENERGIZED (ADJUSTABLE).
 - WEEK THREE: CH1 & P-1 & P-3 SHALL BE IN OPERATION AND CH2 & P-2 SHALL BE DE-ENERGIZED (ADJUSTABLE).
 - WEEK FOUR: CH2 & P-1 & P-2 SHALL BE IN OPERATION AND CH1 & P-3 SHALL BE DE-ENERGIZED (ADJUSTABLE).
 - WEEK FIVE: CH1 & P-2 & P-3 SHALL BE IN OPERATION AND CH2 & P-1 SHALL BE DE-ENERGIZED (ADJUSTABLE).
 - WEEK SIX: CH2 & P-1 & P-3 SHALL BE IN OPERATION AND CH1 & P-2 SHALL BE DE-ENERGIZED (ADJUSTABLE).ROTATION SHALL BE INITIATED BASED ON AN OPERATOR ENTERED DAY INTERVAL OR BY THE CYCLING OF A BINARY POINT.
- THE METHOD OF CYCLING SHALL BE OPERATOR SELECTABLE. CHILLER MODULES MAYBE FORCED INTO A NEW ROTATION SEQUENCE BY CYCLING CHILLER MODULES AT THE TIME OF INITIATION. ALTERNATELY CHILLER MODULE CYCLING CAUSED BY CHILLER MODULE OR PUMP FAILURE SHALL ELIMINATE CHILLER MODULE CYCLING UNTIL FAILURE IS FIXED. CHILLER SOFT START - THE CHILLER SEQUENCING SOFTWARE WILL PROVIDE A USER ADJUSTABLE LOADING TIME AT SYSTEM START-UP.
- WHENEVER THERE IS A REQUIREMENT FOR CHILLED WATER, THE CHILLED WATER SYSTEM SHALL BE INDEXED TO OPERATE.
- UPON A REQUEST FROM THE DDC SYSTEM, THE OPERATIONAL CHILLER MODULE AND PUMPS FOR THAT WEEK (ADJUSTABLE) SHALL BE ENERGIZED.
- THE ASSOCIATED PUMP SPEED SHALL BE MODULATED THROUGH A PID ALGORITHM TO MAINTAIN AN ADJUSTABLE PRESSURE DIFFERENTIAL SENSED AT THE MAIN CHILLED WATER PIPING ADJACENT TO THE MOST REMOTE AHU. THE PUMP VFD SHALL ALWAYS START IN AN UNLOADED CONDITION, AND SLOWLY RAMP UP TO ITS REQUIRED OPERATING SPEED. DP SETPOINT SHALL AUTOMATICALLY ADJUST BASED ON CHILLED WATER VALVE POSITION. 95% OF ALL CHW VALVES SHALL BE CALLING FOR LESS THAN 100% OPEN.
- ONCE THE PUMP AND CHILLER MODULE PROVE FLOW VIA DIFFERENTIAL PRESSURE TRANSMITTERS, THE CHILLER SHALL BE RELEASED TO SEQUENCE. CONTINUOUS DIFFERENTIAL PRESSURE DIFFERENTIAL AND PUMP DIFFERENTIAL PRESSURE TRANSMITTERS. CHILLER MODULES SHALL NOT BE CAPABLE OF OPERATING WITHOUT PROOF OF FLOW.
- UPON A REQUEST FOR THE LEAD CHILLER MODULE, THE DDC SYSTEM SHALL ENERGIZE THE LEAD CHILLED WATER PUMP. UPON PROOF OF PRIMARY FLOW, THE DDC SYSTEM SHALL THEN ENERGIZE THE CHILLER THROUGH ITS LOCAL CONTROL PANEL. THE ASSOCIATED ISOLATION CONTROL VALVE SHALL SHUT UPON THE CHILLER SHUTTING DOWN.
- AFTER ENABLING THE LOCAL CHILLER PANEL, THE DDC SYSTEM SHALL MONITOR THE CHILLER RUNNING LOAD AMPS FOR PROOF OF OPERATION.
- THE LOCAL CHILLER CONTROL PANEL SHALL THEN CYCLE THE CHILLER MODULE'S COMPRESSORS IN SEQUENCE TO MAINTAIN THE EVAPORATOR LEAVING WATER TEMPERATURE AND RUNNING LOAD, AS RESET BY RESET, DEMAND LIMITING AND OTHER APPROPRIATE PROGRAMS.

CONTROLS:

- THE LOCAL CHILLER OPERATION SHALL BE CONTROLLED BY AN INTELLIGENT MICRO-PROCESSOR BASED CONTROLLER IN COMMUNICATION WITH THE DDC SYSTEM VIA SIEMENS APOGEE P2 NETWORK. HARDWIRED INDIVIDUAL POINTS, OR OTHERWISE AS REQUIRED TO BE COMPATIBLE WITH THE CONTROLS INTERFACE(S) OF THE CHILLERS FURNISHED.
- ALL OPERATIONAL DATA SHALL READ-OUT, AND BE OPERATOR MODIFIABLE THROUGH THE DDC OPERATIONS STATION.
- PROVIDE THE STATUS DISPLAY REPORTS, AND ALARM PARAMETERS AS SCHEDULED HEREIN.
- THE CHILLER PLANT CONTROL PROGRAM SHALL BE FULLY EDITABLE AND SET-UP VIA POINT AND CLICK ON A STANDARD WINDOWS SCREEN. IT SHALL NOT REQUIRE SPECIAL SOFTWARE TOOLS OR A BAS TECHNICIAN TO OPERATE AND MODIFY CHILLER SEQUENCING CONTROL.
- THE CHILLER SEQUENCING SOFTWARE SHALL PERFORM THE FOLLOWING CONTROL STRATEGIES:
 - CHILLER PLANT SYSTEM SCHEDULING
 - CHILLER SEQUENCING
 - COLOR GRAPHIC BASED CHILLER PLANT STATUS SCREENS
 - COLOR GRAPHIC BASED CHILLER STATUS SCREENS
 - SYSTEM AND CHILLER DIAGNOSTIC MESSAGES
 - SYSTEM AND CHILLER REPORTS
- CHILLER SEQUENCING - THE CHILLER SEQUENCING SOFTWARE WILL START AND STOP SYSTEM WATER PUMPS AND CHILLERS BASED UPON SYSTEM LOAD.
 - WHEN THE CHILLED WATER SYSTEM IS ENABLED, THE CHILLER SYSTEM CONTROL WILL START THE SYSTEM CHILLED WATER PUMP AND PROVE FLOW THROUGH THE EVAPORATOR.
- ADDITIONAL CHILLER SEQUENCING FOR VARIABLE PRIMARY FLOW CHILLED WATER SYSTEMS:
 - THE SYSTEM CHILLED WATER SUPPLY SETPOINT SHALL BE 34°F AND EDITABLE BY THE OPERATOR. ALTERNATELY THE SYSTEM SETPOINT SHALL BE SET BY THE CHILLER SEQUENCING SOFTWARE PROGRAM.
 - THE CHILLER SEQUENCING SOFTWARE SHALL CONTROL INDIVIDUAL CHILLER SETPOINTS TO MAINTAIN THE SYSTEM SUPPLY TEMPERATURE AT SETPOINT. CHILLER SEQUENCING SHALL FOLLOW SCHEDULE ABOVE.
 - THE CHILLER SEQUENCING SOFTWARE SHALL CONSIDER STOPPING THE CHILLER WHENEVER THERE IS NO CHILLED WATER SYSTEM DEMAND.
 - THE CHILLER SEQUENCING SOFTWARE SHALL NOT SHUTDOWN A CHILLER PUMP UNTIL ALL CHILLER COMPRESSORS ARE PROVEN OFF.
 - THE CHILLER SHALL BE ABLE TO CALL FOR ITS EVAPORATOR TO RUN AT ANY TIME TO PREVENT EVAPORATOR FREEZE-UP WHENEVER THE AMBIENT TEMPERATURE IS 40°F OR LOWER.
 - CHILLER MINIMUM FLOW BY-PASS VALVE CONTROL.
 - THE CHILLER MINIMUM FLOW BY-PASS LINE AND VALVE SHALL BE SIZED TO ALLOW FOR THE MANUFACTURER'S RECOMMENDED MINIMUM FLOW, WITH ALL LOAD CONTROL VALVES CLOSED.
 - THE CHILLER MINIMUM FLOW BY-PASS VALVE SHALL BE A NORMALLY OPEN VALVE.
 - THE CHILLER MINIMUM FLOW BY-PASS VALVE SHALL BE MODULATED TO THE FULLY OPEN POSITION WHEN THE SYSTEM IS SHUTDOWN. THIS SHALL BE DONE TO PREVENT WATER HAMMER WHEN A PUMP IS STARTED AND TO ALLOW FOR MINIMUM FLOW IN THE EVENT THE CHILLER CALLS FOR PUMP OPERATION.
 - FOLLOWING THE CONFIRMED START OF THE LEAD CHILLER MODULE AND WHENEVER SYSTEM IS ENABLED CHILLER SEQUENCING SYSTEM SHALL MODULATE THE CHILLER MINIMUM FLOW BY-PASS VALVE SUCH THAT THE CHILLED WATER FLOW THROUGH ANY OPERATING CHILLER MODULE SHALL NOT DROP BELOW THE MANUFACTURER'S RECOMMENDED MINIMUM FLOW.
 - THE CHILLER OPERATING FLOW SHALL BE DETERMINED BY DIRECT MEASUREMENT USING DPT SENSORS. THE SETPOINT SHALL BE DETERMINED BASED ON THE MANUFACTURER'S RECOMMENDED MINIMUM FLOW RATE AND MEASUREMENTS BY THE T&B CONTRACTOR USING AND ULTRASONIC FLOW METER. UPON SENSING A CHILLER MODULE FAILURE THE CHILLER SEQUENCING SOFTWARE SHALL LOCKOUT THAT CHILLER AND PUMP AND IMMEDIATELY INITIATE THE START OF THE NEXT CHILLER IN THE ROTATION SEQUENCE.

CHILLER SYSTEM OPERATOR INTERFACE:

- CHILLER PLANT CONTROL APPLICATION OPERATIONAL STATUS SCREEN TO INCLUDE:
 - CHILLER SYSTEM STATUS (OFF/SOFT START/NORMAL/AMBIENT LOCKOUT/SHUTDOWN IN PROGRESS)
 - CHILLER PLANT SUPPLY WATER SETPOINT
 - CHILLED WATER SYSTEM SUPPLY WATER TEMPERATURE
 - CHILLED WATER SYSTEM RETURN WATER TEMPERATURE
 - INDIVIDUAL CHILLER FAILURE RESET (PUSH BUTTON)
 - ALL CHILLER FAILURE RESET (PUSH BUTTON)
 - SYSTEM PUMP FAILURE RESET (PUSH BUTTON)
 - MANUAL CHILLER SHUTDOWN (PUSH BUTTON)
 - MANUAL ROTATION OF CHILLER SEQUENCE (PUSH BUTTON)
- CHILLER PLANT CONTROL PROGRAM SHALL INCLUDE A SCREEN(S) THAT ALLOWS EDITING OF THE FOLLOWING DATA WITHOUT ENTERING PROGRAM CODE EDITOR:
 - SUPPLY WATER SETPOINT
 - SYSTEM SOFT LOADING SETPOINTS
 - AMBIENT LOCKOUT SETPOINTS
 - CHILLER SHUTDOWN SETPOINTS
 - AUTO ROTATION SETPOINTS
 - ALARM HANDLING SETUP
 - SECURITY SETUP
- INDIVIDUAL CHILLER STATUS GRAPHIC SCREEN(S)
 - CHILLER NAME
 - CHILLER OPERATING MODE
 - CHILLED WATER SETPOINT
 - CHILLER FLA %
 - CHILLER NUMBER OF STARTS
 - CHILLER RUN HOURS
 - ENTERING CHILLER WATER TEMPERATURE
 - LEAVING CHILLED WATER TEMPERATURE
 - EVAPORATOR FLOW STATUS
 - PURGE UNIT STATUS (LOW PRESSURE MACHINES ONLY)
 - PURGE COMPRESSOR DAILY RUN TIME (LOW PRESSURE MACHINES ONLY)
 - COMPRESSOR PHASE AMPERAGES %
 - COMPRESSOR PHASE AMPERAGES % OF FLA 1/2/3

CHILLER STATUS REPORT:

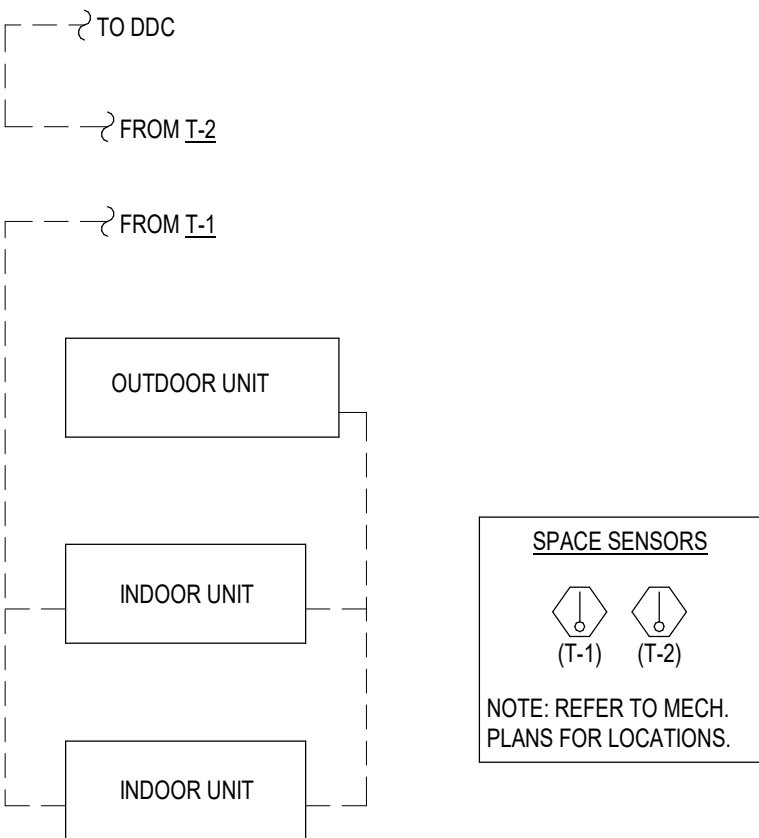
- PROVIDE AN OPERATING STATUS REPORT FOR EACH CHILLER. THE REPORT(S) SHALL PROVIDE THE PRESENT STATUS FOR THE FOLLOWING INFORMATION TO PROVIDE THE OPERATOR WITH CRITICAL CHILLER OPERATING DATA.
 - COMPRESSOR ON/OFF STATUS
 - COMPRESSOR STARTS/RUN HOURS
 - COMPRESSOR PHASE 1/2/3 PERCENT FLA - SEPARATE FOR EACH COMPRESSOR
 - COMPRESSOR CURRENT DRAW - FLA PERCENT
 - ACTIVE CHILLER DIAGNOSTICS OR ALARMS
 - LEAVING CHILLED WATER TEMPERATURE
 - ENTERING CHILLED WATER TEMPERATURE
 - CHILLED WATER SETPOINT
 - REFRIGERANT TEMPERATURE EVAPORATOR/CONDENSER - SEPARATE FOR EACH CIRCUIT
 - OPERATING MODE
 - CHILLER MODEL AND SERIAL NUMBER
 - CHILLED WATER CHILLER LOAD TRENDS: THE DDC SYSTEM SHALL CONTINUOUSLY MEASURE THE CHILLED WATER SUPPLY FLOW RATE, AND THE CHWIS AND CHWR TEMPERATURES, AND CALCULATE THE TOTAL BUILDING COOLING LOAD USING THE FORMULA BELOW:
 - BUILDING LOAD (BTU/H) = FLOW (GPM) X ((CHWS TEMP - CHWR TEMP) (DEG F) X 500).

DIAGNOSTICS/PROTECTION:

- THE DDC SYSTEM SHALL BE ABLE TO ALARM FROM ALL SENSED POINTS AND DIAGNOSTIC ALARMS SENSED BY THE CHILLER CONTROLLER.
 - ALARM LIMITS SHALL BE DESIGNATED FOR ALL SENSED POINTS.
 - SENSED DIAGNOSTIC POINTS SHALL INCLUDE THE FOLLOWING AT A MINIMUM:
 - SYSTEM CHILLED WATER SUPPLY TEMPERATURE
 - SYSTEM CHILLED WATER RETURN TEMPERATURE

PRIMARY VARIABLE SPEED CHILLED WATER CIRCULATION PUMPS STAGING CONTROL:

- VARIABLE CHILLED WATER PUMP CONTROL: THE DDC SYSTEM SHALL CONTROL THE SPEED OF THE LEAD PRIMARY CHILLED WATER PUMP THROUGH THE VFD TO MAINTAIN 10 PSIG (ADJUSTABLE) DIFFERENTIAL PRESSURE AT THE DP TRANSMITTER LOCATED AT THE END OF THE CHILLED WATER LOOP. DP SETPOINT SHALL AUTOMATICALLY ADJUST TO MAINTAIN CHW VALVE POSITION (95% OF ALL CHILLED WATER VALVES SHALL BE LESS THAN 100% OPEN). UNDER NO CIRCUMSTANCE, HOWEVER, WILL THE VARIABLE FREQUENCY DRIVE APPLIED TO CHILLED WATER PUMP MOTOR BE ALLOWED TO EXCEED THE PUMP MOTOR FLA (FULL LOAD AMPS) AT ANY TIME. EACH VFD SHALL BE PROGRAMMED TO ACHIEVE THE FLA OF EACH MOTOR. ON A FAILURE OF THE PRIMARY PUMP DEDICATED TO A CHILLER, THE DDC SYSTEM SHALL AUTOMATICALLY SHUTDOWN ASSOCIATED CHILLER AND BRING ON THE NEXT SEQUENCED PUMP AND CHILLER AND SHALL CONTROL THE SPEED OF THE PUMP TO MAINTAIN THE DIFFERENTIAL PRESSURE AT THE END OF THE CHILLED WATER LOOP, AND SEND AN ALARM OF THE FAILURE.
- UNDER PROGRAM CONTROL, EACH CHILLED WATER PUMP VFD WILL BE AUTOMATICALLY STARTED AND STOPPED VIA A DDC COMMUNICATION BUS WITH AN ASSOCIATED DDC AO. A CURRENT SWITCH, INSTALLED AT EACH PUMP VFD, WILL PROVIDE ON/OFF STATUS TO THE DDC AT ALL TIMES.
- BASED UPON PUMP SPEED, PUMPS WILL BE STAGED ON AND OFF BY DDC TO ACCOMMODATE THE BUILDING FLOW RATE REQUIREMENTS UNDER THE FOLLOWING DDC OPERATIONAL SEQUENCES: UNDER AUTOMATIC CONTROL, SOFTWARE SUBROUTINES, USING "FIFTEEN" LOGIC STATEMENTS, WILL PROVIDE FOR "ON/OFF" STAGING AS DESCRIBED.
- IF DDC SHOULD ATTEMPT TO AUTOMATICALLY START A PUMP AND THE ASSOCIATED SENSOR CAN NOT PROVIDE A SUCCESSFUL START, AN ALARM WILL BE ROUTED TO OPERATOR AT THE OWS (OPERATOR WORK STATION) INDICATING IMMEDIATE PUMP ATTENTION IS NECESSARY AND THE NEXT OPERATIONAL SEQUENCE WILL THEN IMMEDIATELY BE ACTIVATED.
- IF DDC SHOULD ATTEMPT TO AUTOMATICALLY STOP A PUMP AND THE ASSOCIATED CHILLER CAN NOT PROVIDE A SUCCESSFUL STOP, AN ALARM WILL BE ROUTED TO OPERATOR AT THE OWS INDICATING IMMEDIATE PUMP ATTENTION IS NECESSARY.



SEQUENCE OF OPERATION

GENERAL:

- THE MINI-SPLIT SYSTEM SERVES AS COOLING FOR THE MECHANICAL ROOM AND IS COMPRISED WITH A DX COOLING COIL AND CIRCULATION FAN.
 - SPACE TEMPERATURE SET POINT:
 - COOLING 80°F (ADJ.)
 - HEATING 68°F (ADJ.)
- STARTUP/SHUTDOWN:
- UNIT SHALL BE THERMOSTATICALLY CONTROLLED THROUGH THE INTEGRAL MANUFACTURER CONTROLS TO CYCLE WITH LOADS. THERMOSTAT T-1 SHALL BE PROVIDED WITH INDOOR UNIT AND SHALL BE TIED INTO OUTDOOR INDOOR CONTROLLER.
 - ALARM SHALL BE SENT TO DDC SYSTEM UPON RISE IN TEMPERATURE IN THE SPACE GREATER THAN 80°F (ADJ.) OR LOWER THAN 40°F (ADJ.) AS OBSERVED BY T-2.

INPUT/OUTPUT SUMMARY														
SYSTEM	GRAPHIC	INDICATION							ALARM			CONTROL		
		STATUS	TEMPERATURE	RELATIVE HUMIDITY	PPH	CFM	STATIC PRESSURE	DIFF. PRESSURE	GPM	POSITION	RUN TIME	HERTZ	LOW	HIGH
IDU-1 / ODU-1														
SYSTEM														
SPACE (MANU.)														
SPACE														

CONTROL DIAGRAM - AIR COOLED CHILLERS
N.T.S.

DUCTLESS MINI SPLIT CONTROL DIAGRAM
N.T.S.

Revisions

Date

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ARCHITECT/ENGINEER OF RECORD

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Office of Construction and Facilities Management

U.S. Department of Veterans Affairs

Drawing Title

MECHANICAL CONTROLS

Approved:

Phase

FINAL CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title

VAMC Durham - Replace Chillers for AHU 1 & 2

DURHAM VA

Issue Date

03/26/2026

Checked

JMC

Drawn

AJZ

Project Number

V/A Project No. 558-22-172

AG Project No. 0122096

Building Number

1

Drawing Number

M1701

VA FORM 08 - 6231

1

2

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Project Number	
VA Project No.	558-22-172
AG Project No.	0122096
Building Number	
Drawing Number	
M-801	

VA FORM 08-6231

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AutoCAD Docs\0122096 - VAMC Durham - Replace Chillers for AHU 1 & 2\0122096 - VAMC Durham - Replace Chillers - R25_MEPFFAT.rvt

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EQUIPMENT SCHEDULE													
IDENTIFICATION	EQUIPMENT NAME	HP	FLA	MCA	VOLT / PH / LOAD (VA)	MOCp	CB	AMPS	PANEL	CKT #	FEEDER SIZE	CONDUIT	MOTOR STARTING
CH-1	AIR COOLED WATER CHILLER 1	-	-	294	480 V/3-16200 VA	400	400		EX. MCC-N	1	2 SETS OF (3#3/4"AWG + 1#3/4"AWG GND)	(2) 2"	VFD
CH-2	AIR COOLED WATER CHILLER 2	-	-	294	480 V/3-16200 VA	400	400		EX. MCC-N	2	2 SETS OF (3#3/4"AWG + 1#3/4"AWG GND)	(2) 2"	VFD
GMU-1	GLYCOL MAKEUP SYSTEM	(2) 1/3	-	-	120 V/1-1728 VA	20	20		LPM	1	2#12AWG + 1#12AWG GND	3/4"	FACTORY CONTROL PANEL
IDU-1	VRF INDOOR UNIT	-	2.9	-	208 V/1-603 VA	15	15		LPM	3.5	2#12AWG + 1#12AWG GND	3/4"	N/A
IDU-2	VRF INDOOR UNIT	-	2.9	-	208 V/1-603 VA	15	15		LPM	7.9	2#12AWG + 1#12AWG GND	3/4"	N/A
ODU-1	VRF OUTDOOR UNIT	-	-	V	208 V/15.185 VA	40	40		LPM	11.13	2#8AWG + 1#10AWG GND	3/4"	N/A
P-1	CHILLED WATER PUMP	25	-	-	480 V/3-26267 VA	-	50		HPM	1.33	3#8AWG + 1#10AWG GND	1"	VFD
P-2	CHILLED WATER PUMP	25	-	-	480 V/3-26267 VA	-	50		HPM	2.48	3#8AWG + 1#10AWG GND	1"	VFD
P-3	CHILLED WATER PUMP	25	-	-	480 V/3-0 VA	-	50		HPM	7.9.11	3#8AWG + 1#10AWG GND	1"	VFD

DISCONNECT SCHEDULE							
DISC NAME	TYPE OF EQUIPMENT	VOLTAGE RATING	ENCLOSURE RATING	DISC AMP/POLE	FUSEBREAKER SIZE	STARTER SIZE	DISCONNECT NOTES
DS ODU-1	DISCONNECT SWITCH	600V	NEMA 3R	60A/3P	N/A	N/A	-
TS IDU-1	TOGGLE SWITCH	277V	NEMA 12	20A/2P	N/A	N/A	-
TS IDU-2	TOGGLE SWITCH	277V	NEMA 12	20A/2P	N/A	N/A	-

TRANSFORMER EQUIPMENT SCHEDULE						
NAME	SERVE	SUPPLY FROM	SIZE KVA	PRIMARY VOLTAGE	SECONDARY VOLTAGE	LOCATION
TLPM	LPM	HPM	15 KVA	480V	208Y/120V	DRY TYPE AHU-2 MECHANICAL ROOM

TYPICAL SPACE TYPE		LINE VOLTAGE CONTROL DEVICE										LOW VOLTAGE CONTROL DEVICE					SWITCHING/DIMMING					MISCELLANEOUS					NOTES																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
		TOGGLE SWITCHES	CEM/HWALL MOUNTED OCCUPANCY SENSORS	WALL BOX OCCUPANCY SENSORS	WALL BOX OCCUPANCY SENSORS WITH SINGLE OUTPUT RELAY	LUMINAIRE WITH BUILT-IN OCCUPANCY SENSOR	DIGITAL SWITCHES	CEM/HWALL MOUNTED OCCUPANCY SENSORS	MOCROT PHOTOSENSORS	ASTRONOMICAL TIMELOCK	PRESCA	SWITCHING	STEREODIMMING	FULL RANGE DIMMING	CENTRALIZED ORCEP/AN RETURN CCT CONTROL	VACANCY SENSOR INELIGIBLE/INT (A MINUTES)	LIGHTS MANUAL ON	LIGHTS AUTOMATIC ON TO 10% LIGHT LEVEL UPON OCCUPANCY	TIMELOCK SCHEDULED ON TIME	TIMELOCK SCHEDULED OFF TIME	OCCT SENSOR TO ENERGIZE TIMELOCK DURING OFF HOURS	LIGHTS CONTROLLED BY INTEGRAL PHOTOCELL																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
1	MECHANICAL ROOM - INTERIOR	X	-	-	-	-	-	-	-	-	-	-	X	-	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

1 LIGHTING CONTROL MATRIX

N.T.S.

2 MOTOR CONTROL CENTER MCC-N - DEMO WORK

N.T.S.

BRANCH PANEL: LPM																		
LOCATION: MECHANICAL ROOM						VOLTS: 120/208 Wye				A.I.C. RATING: 22,000								
SUPPLY FROM: TLPM						PHASES: 3				MAINS TYPE: MCB								
MOUNTING: Surface						WIRES: 4				MAINS RATING: 60 A								
ENCLOSURE: Type 1										SERVICE ENTRANCE: NO								
GENERAL NOTES: PROVIDE INTEGRAL SPD.																		
CKT	CIRCUIT DESCRIPTION					NOTE S	TRIP	POLES	A	B	C	POLES	TRIP	NOTE S	CIRCUIT DESCRIPTION	CKT		
1	GMU-1 REC						20 A	1	1728... 60 VA				1	20 A	INTERIOR LIGHTING	2		
3	IDU-1						15 A	2		603 VA 30 VA			1	20 A	EXTERIOR LIGHT	4		
5											0 VA 540 VA		1	20 A	EXTERIOR REC	6		
7	IDU-2						15 A	2	603 VA 540 VA				1	20 A	INTERIOR REC	8		
9										0 VA 150 VA			1	15 A	HEAT TRACE	10		
11	ODU-1						40 A	2	2593... 0 VA		2593... 0 VA		1	20 A	SPARE	12		
13													1	20 A	SPARE	14		
15	SPARE						20 A	1		0 VA 0 VA		0 VA 0 VA		1	20 A	SPARE	16	
17	SPARE						20 A	1				0 VA 0 VA		1	20 A	SPARE	18	
19	SPARE						20 A	1	0 VA 0 VA					1	20 A	SPARE	20	
21	SPARE						20 A	1		0 VA --				1	--	SPACE	22	
23	SPACE						--	1			--	--	--		1	--	SPACE	24
25	SPACE						--	1	--	--					1	--	SPACE	26
27	SPACE						--	1	--	--	--	--			1	--	SPACE	28
29	SPACE						--	1	--	--	--	--			1	--	SPACE	30
TOTAL...						5425 VA	783 VA	3042 VA										
TOTAL...						48 A	7 A	28 A										
LEGEND:																		
LOAD CLASSIFICATION				CONNECTED LOAD		DEMAND FACTOR		ESTIMATED DEMAND		PANEL TOTALS								
Lighting				90 VA		100.00%		90 VA										
Other				6541 VA		100.00%		6541 VA		TOTAL CONNECTED LOAD: 9246 VA								
Receptacle				2672 VA		100.00%		2672 VA		TOTAL EST. DEMAND: 9246 VA								
										TOTAL CONNECTED... 26 A								
										TOTAL EST. DEMAND... 26 A								
BRANCH CIRCUIT NOTES:																		

LIGHTING FIXTURE SCHEDULE									
TYPE	DESCRIPTION	MANUFACTURER	MODEL	VOLTAGE	VA	LAMP TYPE	FIXTURE LUMENS	FIXTURE COLOR TEMP (K)	DRIVER/BALLAST
A	SURFACE MOUNT, W/BATTERY BACKUP	LITHONIA	CLXL36 3750LM SEF WDL WD MVOLT G21 35K 80CRI E10W BAA	120 V	20 VA	LED	3750	3500	10% DIMMING 0-10V
W	EXTERIOR LED MOUNT LIGHT	LITHONIA	WDC2 LED P2 40K 80CRI FTM MVOLT PE DBLXD	120 V	15 VA	LED	600	4000	N/A

- NOTES: (APPLICABLE TO LUMINAIRE SCHEDULE ONLY)
- CATALOG NUMBERS FOR LUMINAIRES INDICATE THE TYPE AND QUALITY OF THE LUMINAIRE TO BE PROVIDED BY THE CONTRACTOR AND ARE GENERAL IN NATURE. THE CATALOG NUMBERS ARE NOT INTENDED TO INDICATE AN ACTUAL ORDER MODEL NUMBER. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE LUMINAIRES WITH THE TYPE OF LED, DRIVER, LENS OR DIFFUSER, AND CONSTRUCTION FEATURES AS INDICATED IN THE SPECIFICATIONS.
 - EMERGENCY FIXTURES AND BATTERY PACKS ARE TO BE CONNECTED TO CONSTANT HOT LEG (UN-SWITCHED) OF CIRCUIT INDICATED.
 - ALL MOUNTING HEIGHTS SHALL BE AS INDICATED ON THE DRAWINGS, LIGHTING FIXTURE SCHEDULE, OR AS DIRECTED BY ARCHITECT OR ENGINEER. MOUNTING HEIGHTS OF WALL MOUNTED LUMINAIRES SHALL BE MEASURED FROM THE FINISHED FLOOR TO THE CENTERLINE OF THE LUMINAIRE. MOUNTING HEIGHTS OF CEILING SUSPENDED LUMINAIRES SHALL BE MEASURED FROM THE FINISHED FLOOR TO THE BOTTOM OF THE LUMINAIRE.

A 30-DAY MAXIMUM LOAD STUDY WAS PERFORMED ON MCC-A TO DETERMINE IF MCC-A HAD THE ELECTRICAL CAPACITY TO ACCOMMODATE THIS PROJECT. AS OF 12/16/2025, THE MAXIMUM AMPERAGE ON MCC-A WAS RECORDED TO BE 380 AMPS. THOUGH SOME EXISTING LOAD WILL BE REMOVED AS PART OF THIS PROJECT, THE TOTAL LOAD, INCLUDING ALL OF THE EXISTING LOAD AND THE PROPOSED NEW LOAD ADDED, IS APPROXIMATELY 1056 AMPS, OR 53% OF THE CAPACITY OF MCC-A. THE LOAD STUDY THEREFORE VALIDATES THE ELECTRICAL DESIGN APPROACH IN THESE DOCUMENTS.

CODED NOTES-DEMO WORK:

- DE-ENERGIZE, DISCONNECT AND REMOVE FEEDERS FOR CHILLERS. REMOVE EXISTING MCC BUCKET AND REPLACE WITH NEW. REFER TO NEW WORK FOR SIZE.
- DE-ENERGIZE, DISCONNECT AND REMOVE FEEDERS FOR CHILLER PUMPS. MCC BUCKET SHALL BE DESIGNATED AS SPARE.

CODED NOTES-NEW WORK:

- PROVIDE NEW 400A CIRCUIT BREAKER/FEEDER BUCKET FOR NEW CHILLER. THE NEW CIRCUIT BREAKER/FEEDER BUCKET WILL FIT IN THE 24" (TOP TO BOTTOM) EXISTING SPACE ALLOCATED. COORDINATE WITH SQUARE D TO PROVIDE APPROPRIATE MCC BUCKET MODIFICATION KIT TO CONVERT TO FEEDER BREAKER BUCKET. EXISTING MCC IS SQUARE D MODEL 6.
- CIRCUIT BREAKER MADE SPARE DURING DEMOLITION WORK. SPARE TO BE UTILIZED FOR EXTERNAL SURGE PROTECTIVE DEVICE (SPD). PROVIDE NEW CONDUIT AND CONDUCTOR 2E-602.
- PROVIDE NEW FEEDER BUCKET TO SERVE NEW PUMP HOUSE. EXISTING SPACE IS ROUGHLY 10" (TOP TO BOTTOM). COORDINATE WITH SQUARE D TO PROVIDE APPROPRIATE MCC BUCKET MODIFICATION KIT TO CONVERT TO FEEDER BREAKER BUCKET. EXISTING MCC IS SQUARE D MODEL 6.

SECTION 1	SECTION 2	SECTION 3	SECTION 4	SECTION 5	SECTION 6
2000A ELECTRONIC TRIP LSC MAIN CIRCUIT BREAKER SECTION	CH #1 400A, 3P ①	CH #2 400A, 3P ①	SPARE 15A, 3P	HOT WATER P-1 15A, 3P	EXHAUST FAN 100A, 3P
				HOT WATER P-2 15A, 3P	TRANSFORMER 30 KVA 50A, 3P
			SPARE 200A, 3P	CHILLED WATER P-6 15A, 3P	SPARE 15A, 3P
				CHILLED WATER P-3 15A, 3P	SPARE 20A, 3P
				CHILLED WATER P-4 15A, 3P	SPARE 20A, 3P
				MORGUE FAN F-1 15A, 3P	PANEL LPH 200A, 3P ③
				SPARE 15A, 3P	SPACE
				EXTERNAL SPD 50A, 3P ②	SPACE
2000A ELECTRONIC TRIP LSC MAIN CIRCUIT BREAKER SECTION	CH #1 400A, 3P ①	CH #2 400A, 3P ①	SPARE 15A, 3P	HOT WATER P-1 15A, 3P	EXHAUST FAN 100A, 3P
				HOT WATER P-2 15A, 3P	TRANSFORMER 30 KVA 50A, 3P
			SPARE 200A, 3P	CHILLED WATER P-6 15A, 3P	SPARE 15A, 3P
				CHILLED WATER P-3 15A, 3P	SPARE 20A, 3P
				CHILLED WATER P-4 15A, 3P	SPARE 20A, 3P
				MORGUE FAN F-1 15A, 3P	PANEL LPH 200A, 3P ③
				SPARE 15A, 3P	SPACE
				EXTERNAL SPD 50A, 3P ②	SPACE

3 MOTOR CONTROL CENTER MCC-N - NEW WORK

N.T.S.

BRANCH PANEL: HPM																											
LOCATION: MECHANICAL ROOM						VOLTS: 480/277 Wye				A.I.C. RATING: 35,000																	
SUPPLY FROM: MCC-N						PHASES: 3				MAINS TYPE: MCB																	
MOUNTING: Surface						WIRES: 4				MAINS RATING: 200 A																	
ENCLOSURE: Type 1										SERVICE ENTRANCE: NO																	
GENERAL NOTES: PROVIDE INTEGRAL SPD.																											
CKT	CIRCUIT DESCRIPTION					NOTE S	TRIP	POLES	A		B		C		POLES	TRIP	NOTE S	CIRCUIT DESCRIPTION					CKT				
1									9422...	9422...														2			
3	P-1						50 A	3			9422...	9422...				3	50 A		P-2					4			
5													9422...	9422...										6			
7																								8			
9	P-3						50 A	3		0 VA	0 VA			0 VA	0 VA		3	50 A		SPARE					10		
11																								12			
13																								14			
15	TLPM						30 A	3	5425...	0 VA		783 VA	0 VA		0 VA	0 VA	1	20 A		SPARE					16		
17													3042...	0 VA			1	20 A		SPARE					18		
19	SPARE						20 A	1	0 VA	0 VA							1	20 A		SPARE					20		
21	SPARE						20 A	1				0 VA	0 VA				1	20 A		SPARE					22		
23	SPARE						20 A	1						0 VA	0 VA	1	20 A		SPARE					24			
25	SPACE						--	1	--	--							1	--		SPACE					26		
27	SPACE						--	1			--	--					1	--		SPACE					28		
29	SPACE						--	1						--	--	1	--			SPACE					30		
									TOTAL...		24263 VA		19628 VA		21872 VA												
									TOTAL...		89 A		71 A		80 A												
LEGEND:																											
LOAD CLASSIFICATION								CONNECTED LOAD				DEMAND FACTOR				ESTIMATED DEMAND				PANEL TOTALS							
Lighting								90 VA				100.00%				90 VA				TOTAL CONNECTED LOAD: 65761 VA							
Motor								56534 VA				71.50%				63601 VA				TOTAL EST. DEMAND: 72827 VA							
Other								6541 VA				100.00%				6541 VA				TOTAL CONNECTED... 79 A							
Receptacle								2672 VA				100.00%				2672 VA				TOTAL EST. DEMAND... 88 A							
BRANCH CIRCUIT NOTES:																											
1. CHILLED WATER PUMPS ARE LEADILAG.																											

